JADAVPUR UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Assignment 6 Report

CISCO Packet Tracer Network Experiments

LAN Construction with Hubs and Switches, Inter-LAN Routing, DHCP Configuration, DNS and Server Services

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Assignment: Assignment 6 (CISCO Packet Tracer Network Experiments)

Date of Submission: November 2026

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1 Overview and Objectives

Cisco Packet Tracer is a comprehensive network simulation and visualization tool developed by Cisco Systems that allows students to design, build, configure, and troubleshoot virtual networks without requiring physical hardware. It supports a wide range of Cisco devices including routers, switches, hubs, and end-host PCs, providing a realistic environment for experimenting with network topologies, IP addressing, routing protocols, and network services.

This assignment explores fundamental networking concepts through seven progressive experiments in Cisco Packet Tracer, covering Layer-2 LAN construction, address resolution, inter-network routing with static routes, and application-layer services (DHCP, DNS, HTTP, FTP).

1.1 Objectives

The primary objectives of this laboratory assignment are:

- Establish direct point-to-point connectivity between two hosts using a crossover cable and verify reachability via ICMP ping.
- Construct a Local Area Network (LAN) using a hub and verify broadcast-domain connectivity between all host pairs.
- Construct a LAN using a Layer-2 switch and observe the dynamic learning behavior of MAC address tables and ARP caches.
- Interconnect hub-based and switch-based LANs via a crossover cable and verify cross-LAN reachability.
- Design and configure a multi-LAN internetwork with two Cisco 1841 routers, static routing, and WAN serial links to enable inter-LAN communication across different IP subnets.
- Configure DHCP servers on individual LANs to dynamically assign IP addresses and default gateways to hosts.
- Deploy a complete LAN with web, FTP, DHCP, and DNS servers, configure DNS records, and verify name-based service access.

2 Question 1: Direct Host-to-Host Connection via Crossover Cable

2.1 Problem Statement

Connect two hosts back-to-back with a crossover cable. Assign IP addresses, and see whether they are able to ping each other.

2.2 Experimental Setup

Two PCs (PC0 and PC1) were placed in the Cisco Packet Tracer workspace and connected directly using a **crossover Ethernet cable**. A crossover cable is required for direct host-to-host connections because both devices are DTE (Data Terminal Equipment) devices that transmit on the same pin pairs. The crossover cable swaps the transmit and receive pairs (pins 1,2 $\leftrightarrow$ pins 3,6), enabling proper communication.

2.2.1 IP Address Configuration

Host	IP Address	Subnet Mask	Interface
PC0	`192.168.1.1`	`255.255.255.0`	FastEthernet0
PC1	`192.168.1.2`	`255.255.255.0`	FastEthernet0

Table 1: IP address assignment for direct host-to-host connection.

2.3 Ping Test Results

The connectivity was verified by issuing ping commands from PC0 to PC1 and vice versa.

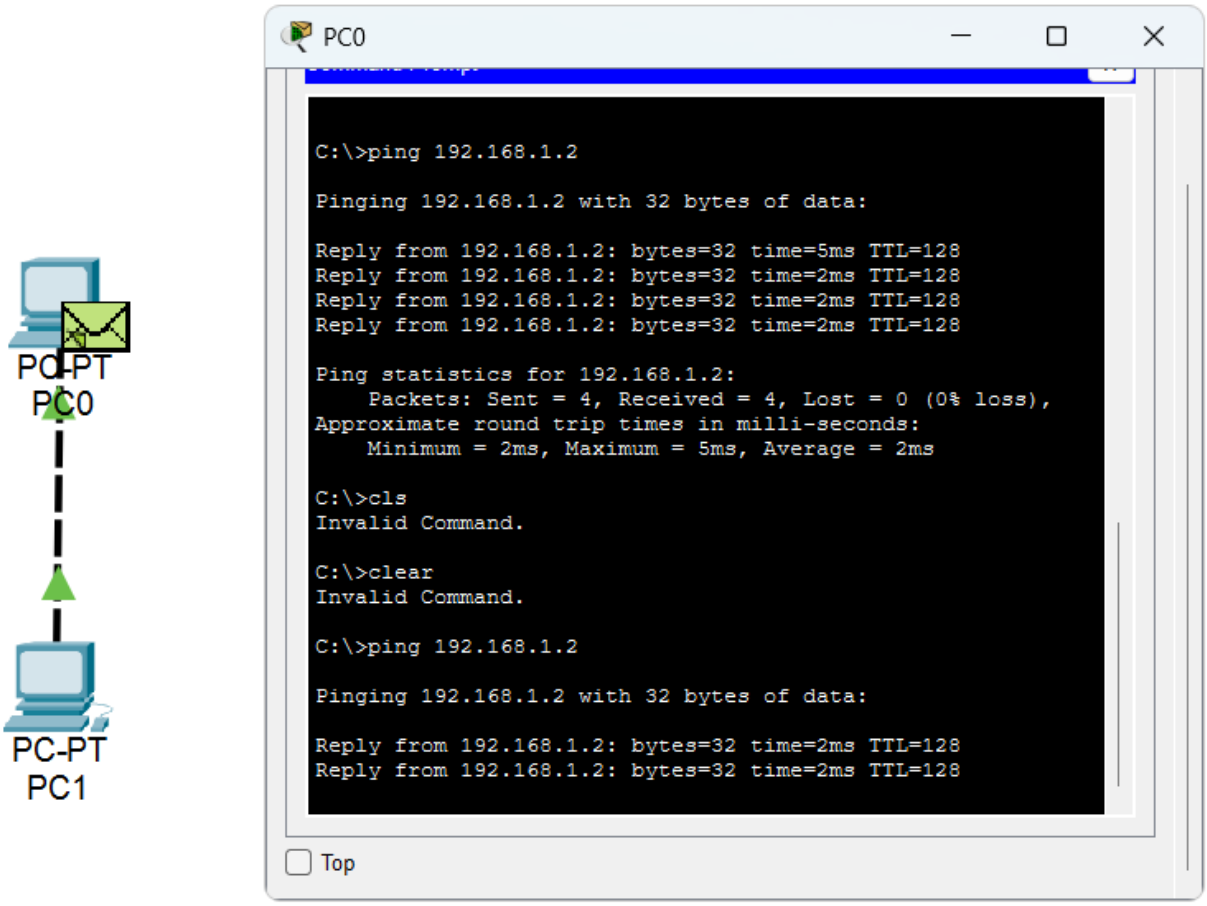


Figure 1: Cisco Packet Tracer showing two PCs connected via crossover cable. PC0 successfully pings PC1 (`192.168.1.2`) with 0% packet loss.

As shown in Figure 1:

- **From PC0:** `ping 192.168.1.2` — All 4 packets received successfully (0% loss). The first ping had a slightly higher latency (5ms) due to the initial ARP resolution, while subsequent pings completed in 2ms.
- **From PC1 (second ping from PC0):** The second ping session also shows all replies received with 2ms round-trip time, confirming stable bidirectional connectivity.

Conclusion: Two hosts connected back-to-back with a crossover cable can successfully communicate when assigned IP addresses on the same subnet. The crossover cable eliminates the need for intermediate networking equipment (hub or switch).

3 Question 2: LAN-A with Three Hosts Using a Hub

3.1 Problem Statement

Create a LAN (named LAN-A) with 3 hosts using a hub. Ping each pair of nodes.

3.2 Experimental Setup

Three PCs (PC0(1), PC1(1), PC2) were connected to a central **Hub-PT (Hub0)** using straight-through Ethernet cables. A hub is a Layer-1 (Physical Layer) device that operates by broadcasting all incoming frames to every connected port, creating a single collision domain.

3.2.1 IP Address Configuration

Host	IP Address	Subnet Mask	Network
PC0(1)	`192.168.2.1`	`255.255.255.0`	LAN-A
PC1(1)	`192.168.2.2`	`255.255.255.0`	LAN-A
PC2	`192.168.2.3`	`255.255.255.0`	LAN-A

Table 2: IP address assignment for LAN-A (hub-based network).

3.3 Ping Test Results

From the Command Prompt on PC0(1), ping tests were conducted to both PC1(1) and PC2:

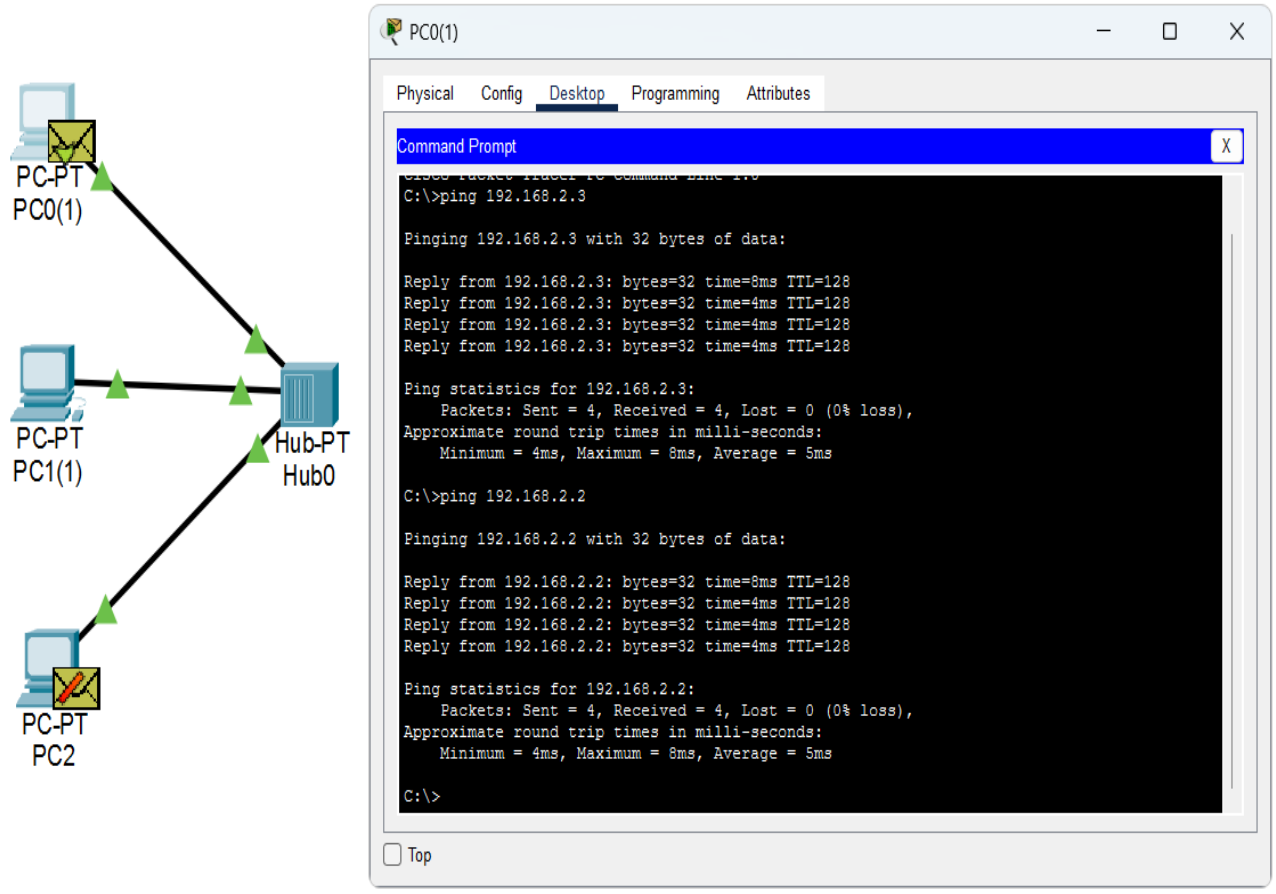


Figure 2: LAN-A topology with three hosts connected via Hub0. PC0(1) successfully pings `192.168.2.3` (PC2) and `192.168.2.2` (PC1(1)) with 0% packet loss.

As shown in Figure 2:

- `ping 192.168.2.3` (PC2): 4/4 packets received, Min=4ms, Max=8ms, Avg=5ms.
- `ping 192.168.2.2` (PC1(1)): 4/4 packets received, Min=4ms, Max=8ms, Avg=5ms.

Observation: All three hosts in LAN-A can communicate with each other through the hub. The hub broadcasts every frame to all ports, so all hosts receive all traffic regardless of the intended destination. This creates a single collision domain but allows connectivity among all connected devices.

4 Question 3: LAN-B with Three Hosts Using a Switch

4.1 Problem Statement

Create a LAN (named LAN-B) with 3 hosts using a switch. Record contents of the ARP Table of end hosts and the MAC Forwarding Table of the switch. Ping each pair of nodes. Now record the contents of the ARP Table of end hosts and the MAC Forwarding Table of the switch again.

4.2 Experimental Setup

Three PCs (PC3, PC4, PC5) were connected to a **Cisco 2960-24TT Switch (Switch0)** using straight-through Ethernet cables. Unlike a hub, a switch is a Layer-2 (Data Link Layer) device that maintains a **MAC Address Table** (also called a CAM table) and forwards frames only to the specific port associated with the destination MAC address.

4.2.1 IP Address Configuration

Host	IP Address	Subnet Mask	Switch Port
PC3	`192.168.3.1`	`255.255.255.0`	Fa0/1
PC4	`192.168.3.2`	`255.255.255.0`	Fa0/2
PC5	`192.168.3.3`	`255.255.255.0`	Fa0/3

Table 3: IP address assignment for LAN-B (switch-based network).

4.3 Network Topology

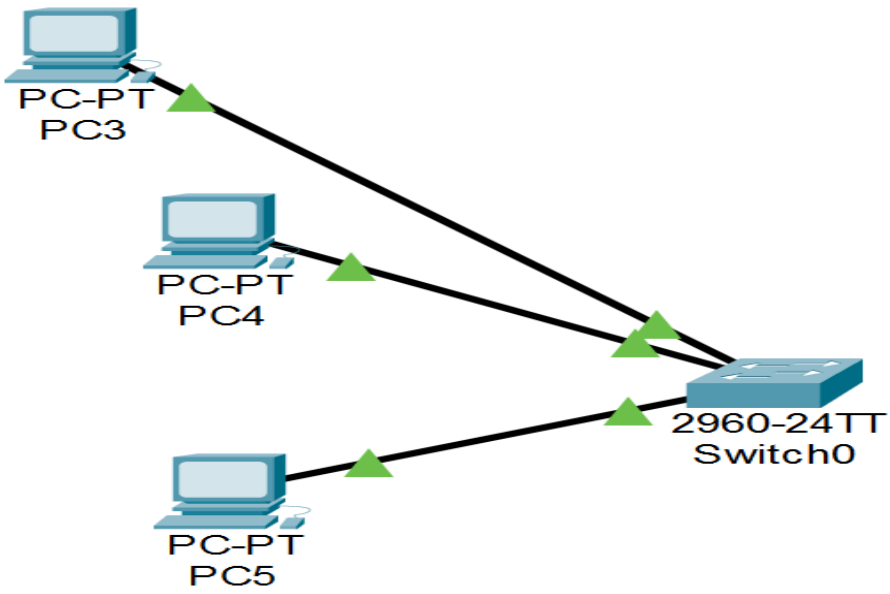
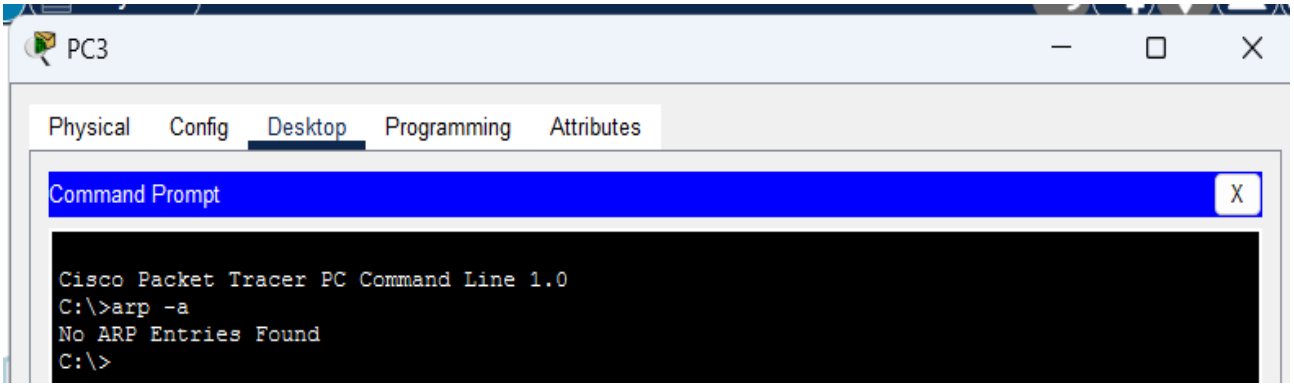


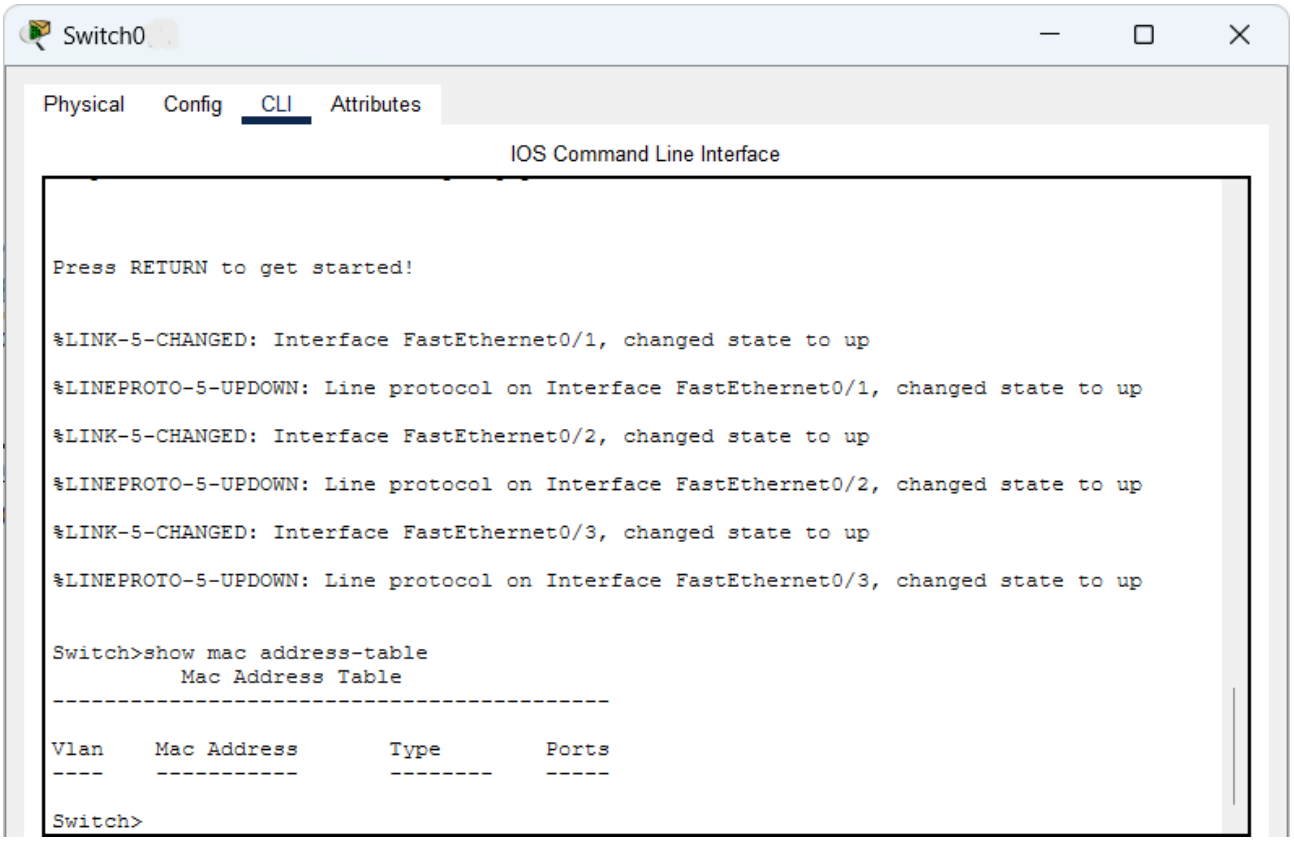
Figure 3: LAN-B topology: Three PCs (PC3, PC4, PC5) connected to a Cisco 2960-24TT switch (Switch0).

4.4 Before Ping: ARP Table and MAC Address Table

Before any communication occurs, the ARP tables on the hosts and the MAC forwarding table on the switch are empty.



(a) PC3: `arp -a` shows "No ARP Entries Found" before any ping.

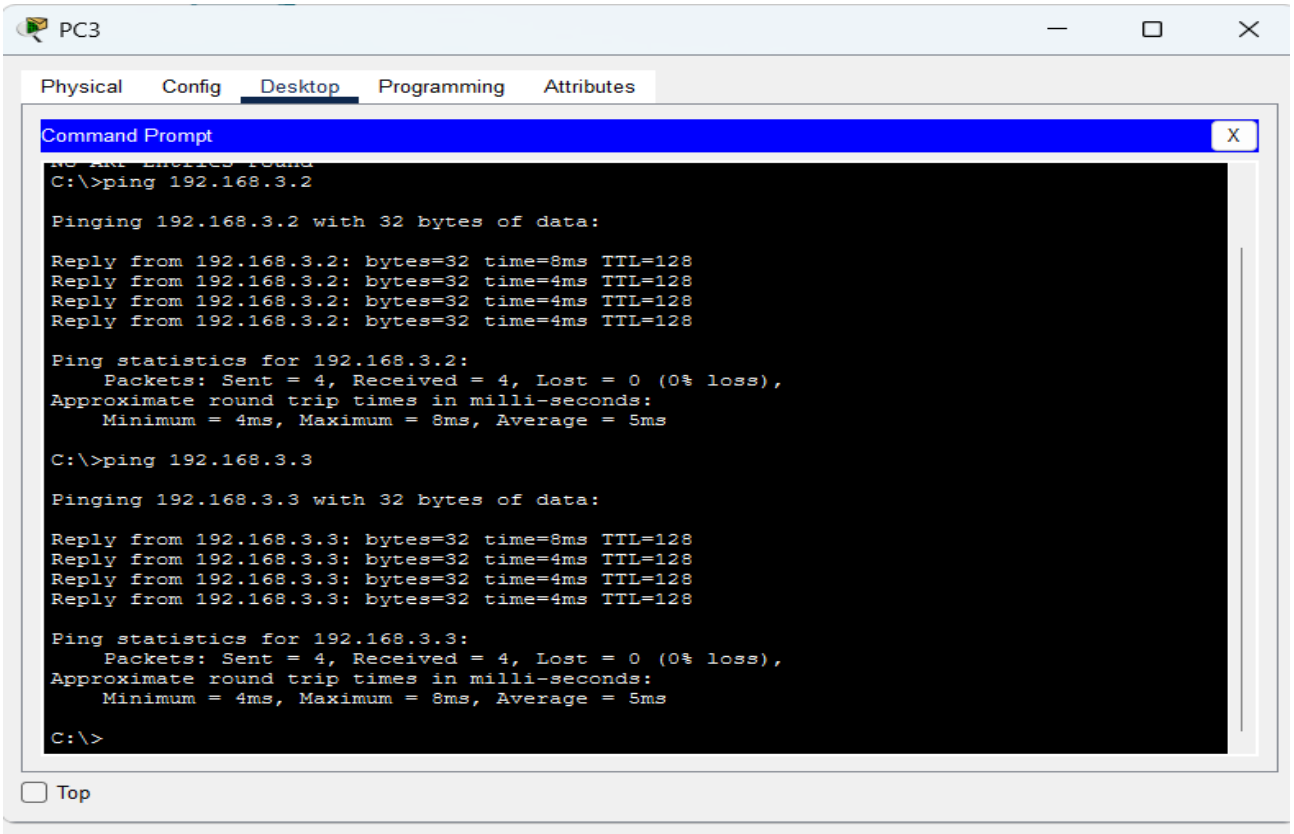


(a) Switch0: `show mac address-table` displays an empty MAC table before any traffic.

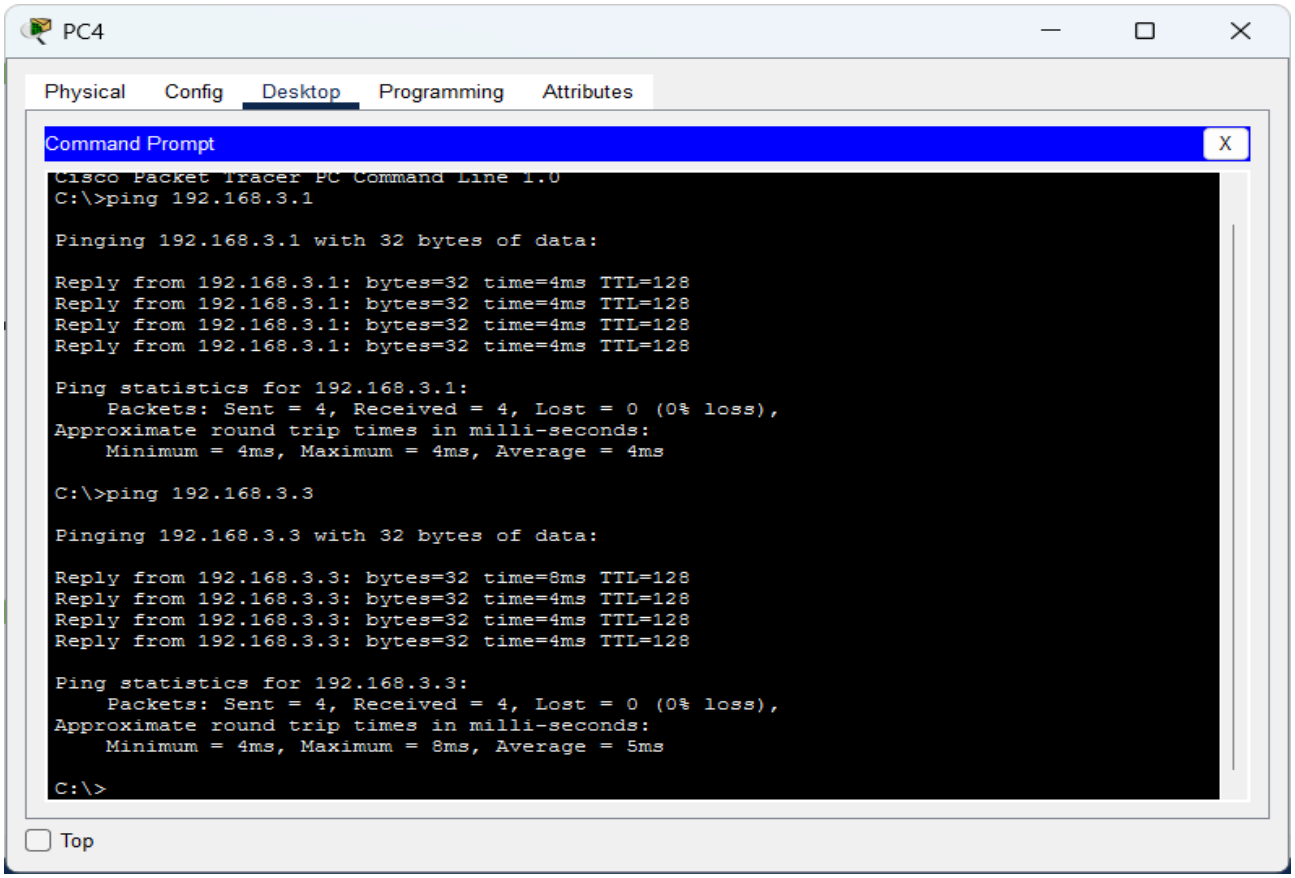
Figure 5: Empty ARP and MAC address tables before any network communication.

4.5 Ping Tests Between All Pairs

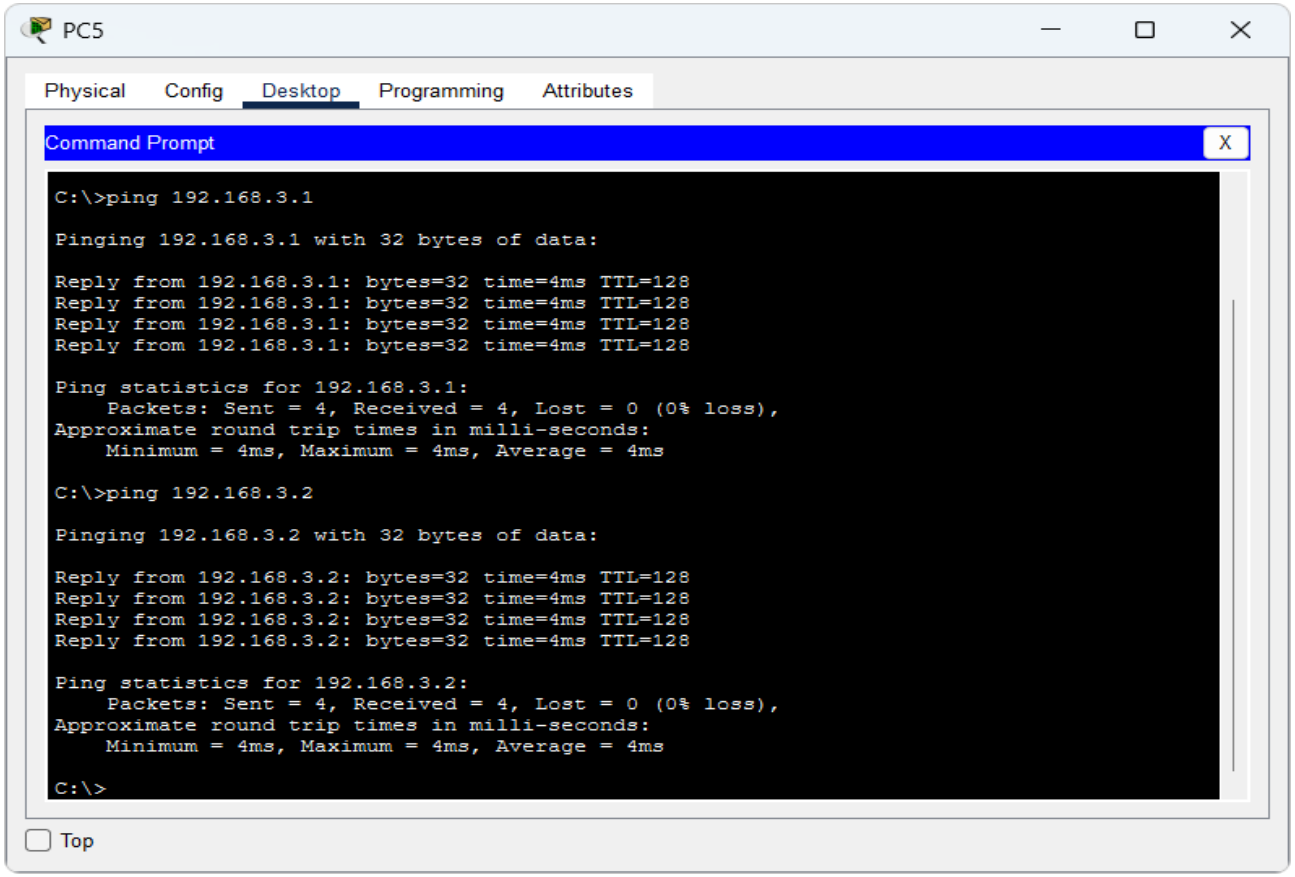
Ping tests were conducted between all three pairs of hosts:



(a) PC3 pings PC4 (`192.168.3.2`) and PC5 (`192.168.3.3`): all successful.



(a) PC4 pings PC3 (`192.168.3.1`) and PC5 (`192.168.3.3`): all successful.

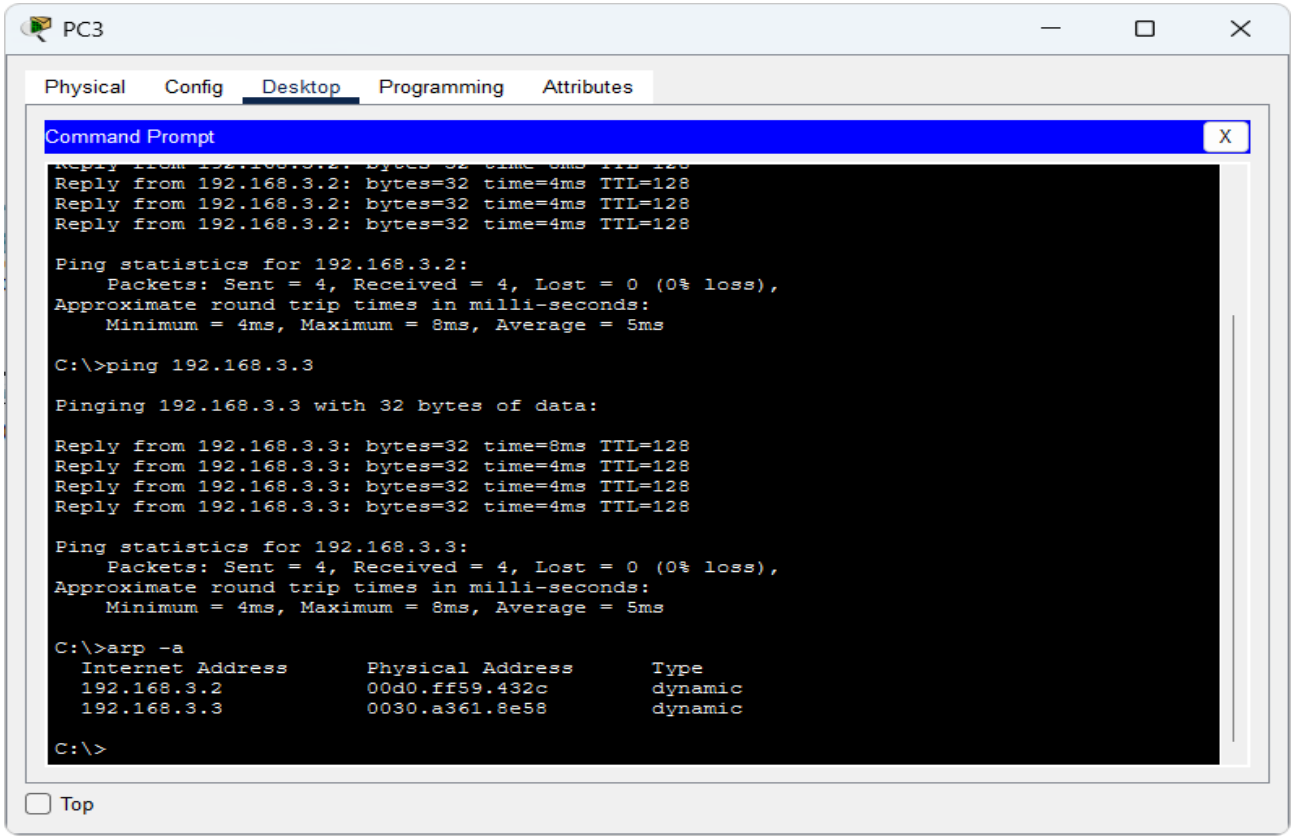


(a) PC5 pings PC3 (`192.168.3.1`) and PC4 (`192.168.3.2`): all successful.

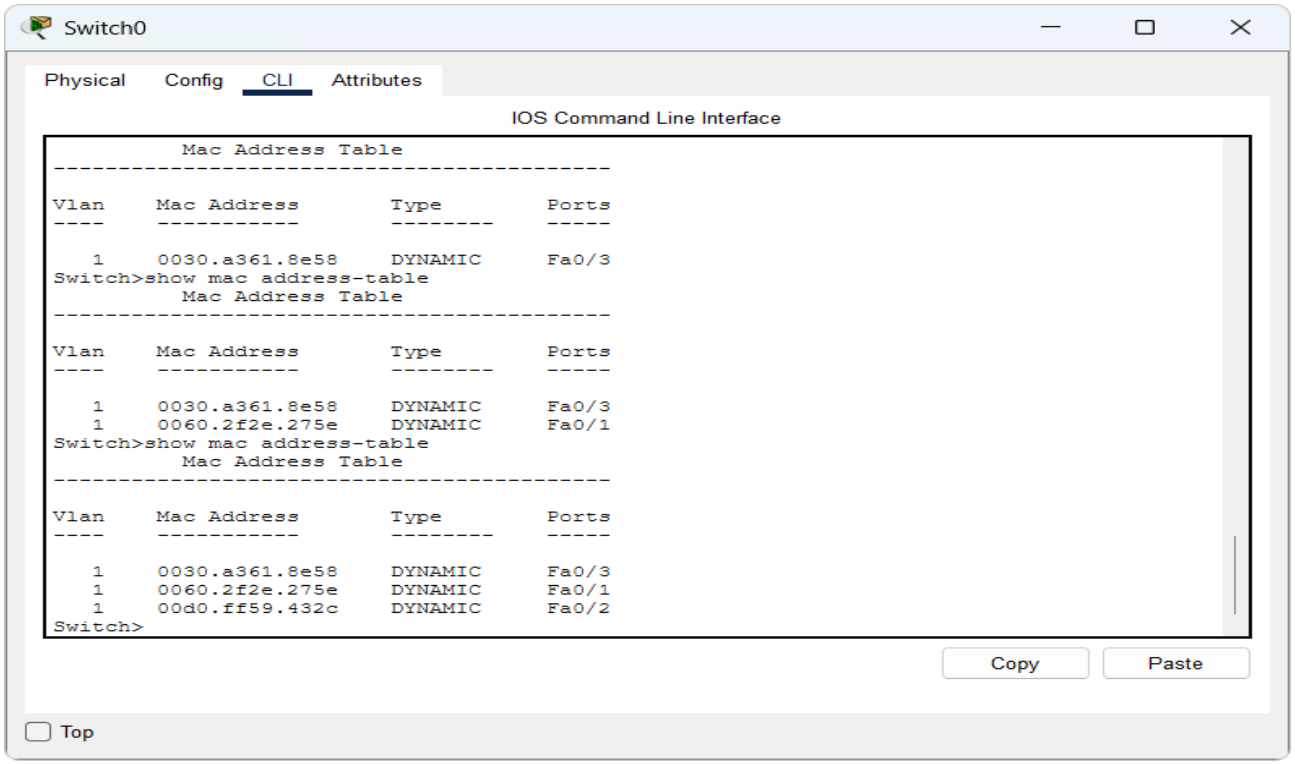
Figure 8: Ping results between all three pairs of hosts in LAN-B.

4.6 After Ping: ARP Table and MAC Address Table

After the ping exchanges, both the host ARP caches and the switch's MAC address table have been dynamically populated.



(a) PC3: `arp -a` now shows learned MAC-to-IP mappings for PC4 and PC5.



(a) Switch0: MAC address table now contains entries for all three connected hosts on their respective ports.

Figure 10: Populated ARP and MAC address tables after ping exchanges.

4.6.1 ARP Table of PC3 (After Pinging)

Internet Address	Physical Address	Type
192.168.3.2	00d0.ff59.432c	dynamic
192.168.3.3	0030.a361.8e58	dynamic

Table 4: ARP table of PC3 after pinging PC4 and PC5.

4.6.2 MAC Address Table of Switch0 (After Pinging)

The switch MAC address table was observed at three progressive stages using `show mac address-table`:

Stage	VLAN	MAC Address	Type	Port
After 1st ping	1	0030.a361.8e58	DYNAMIC	Fa0/3
After 2nd ping	1	0030.a361.8e58	DYNAMIC	Fa0/3
	1	0060.2f2e.275e	DYNAMIC	Fa0/1
After all pings	1	0030.a361.8e58	DYNAMIC	Fa0/3
	1	0060.2f2e.275e	DYNAMIC	Fa0/1
	1	00d0.ff59.432c	DYNAMIC	Fa0/2

Table 5: Progressive MAC address table learning on Switch0.

Key Observation: The switch dynamically *learns* MAC addresses by examining the source MAC address field of incoming Ethernet frames and associates each address with the port on which the frame arrived. This is the fundamental operation of a Layer-2 switch, enabling it to forward frames intelligently rather than broadcasting them to all ports like a hub.

5 Question 4: Connecting LAN-A and LAN-B via Crossover Cable

5.1 Problem Statement

Connect LAN-A and LAN-B by connecting the hub and switch using a crossover cable. Ping between each pair of hosts of LAN-A and LAN-B. Now record the contents of the ARP Table of end hosts and the MAC Forwarding Table of the switch again.

5.2 Experimental Setup

LAN-A (hub-based, 192.168.2.0/24 subnet) and LAN-B (switch-based, originally 192.168.3.0/24) were reconfigured onto the **same subnet** (`192.168.2.0/24`) so that all six hosts could communicate at Layer 2. The hub (Hub1) and switch (Switch2) were interconnected using a **crossover cable**.

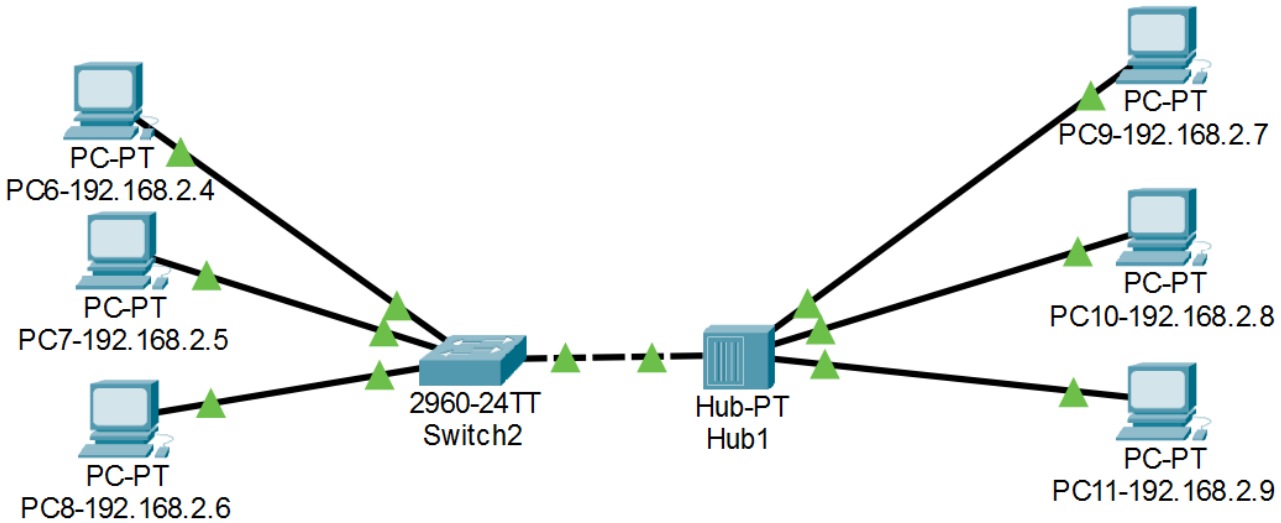


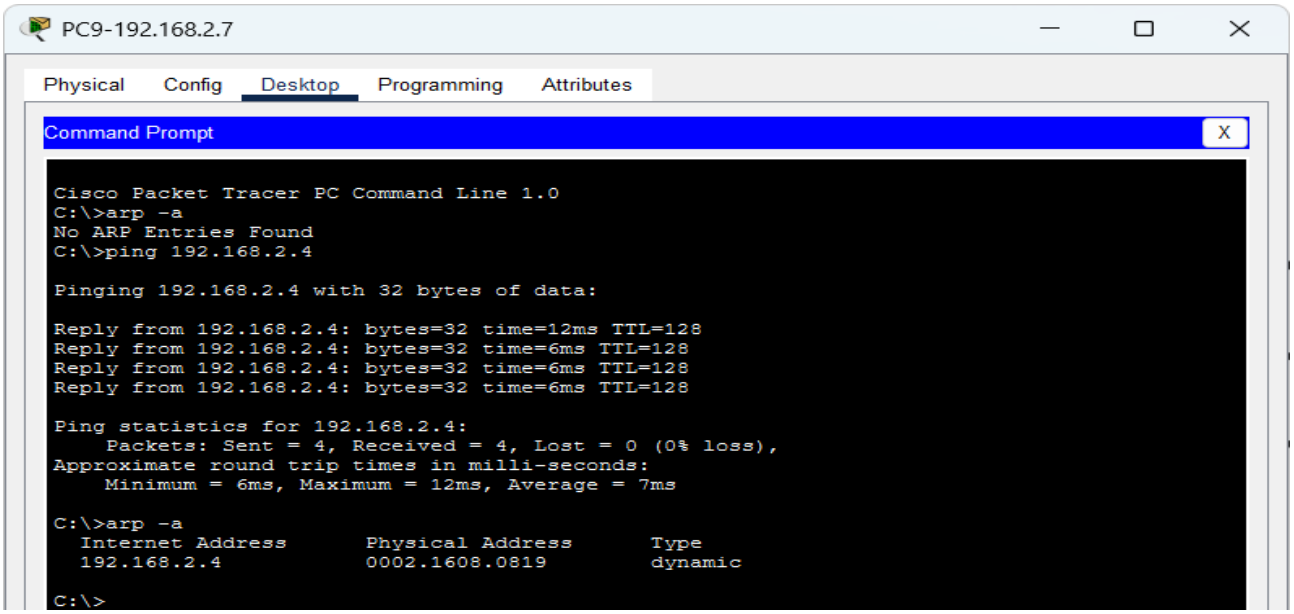
Figure 11: Combined topology: LAN-B hosts (PC6, PC7, PC8 on Switch2) connected to LAN-A hosts (PC9, PC10, PC11 on Hub1) via a crossover cable between the switch and hub. All hosts are on subnet `192.168.2.0/24`.

5.2.1 IP Address Configuration

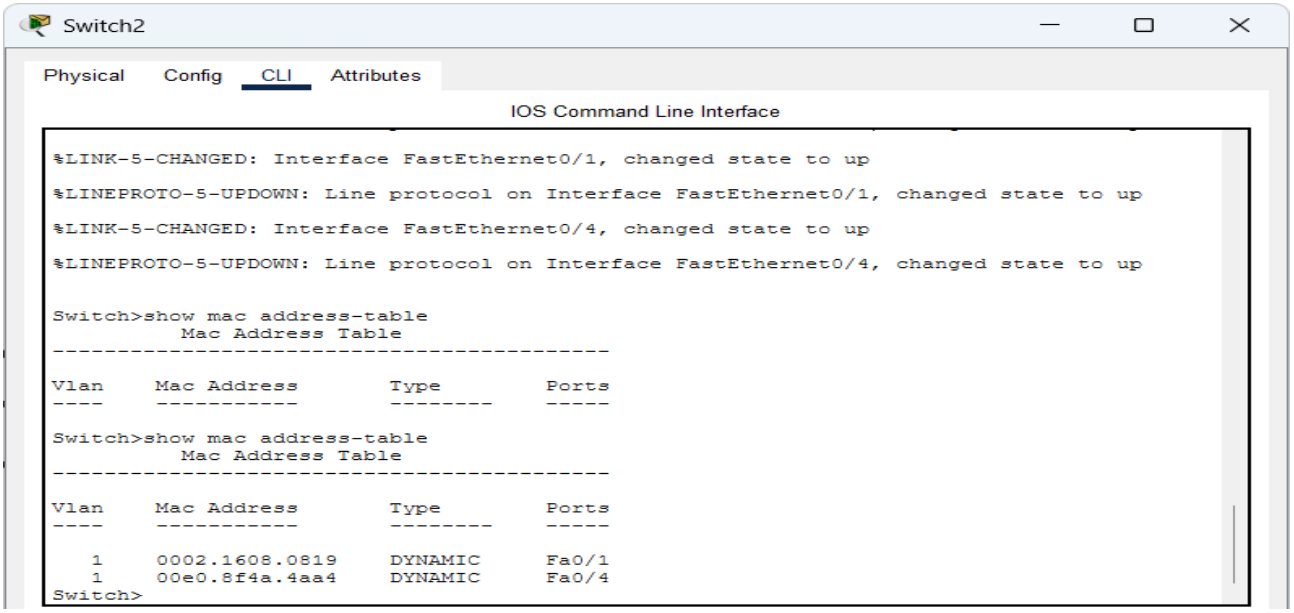
Host	IP Address	Subnet Mask	Connected To
PC6	192.168.2.4	255.255.255.0	Switch2 (LAN-B)
PC7	192.168.2.5	255.255.255.0	Switch2 (LAN-B)
PC8	192.168.2.6	255.255.255.0	Switch2 (LAN-B)
PC9	192.168.2.7	255.255.255.0	Hub1 (LAN-A)
PC10	192.168.2.8	255.255.255.0	Hub1 (LAN-A)
PC11	192.168.2.9	255.255.255.0	Hub1 (LAN-A)

Table 6: IP address assignment for the combined LAN-A and LAN-B network.

5.3 Cross-LAN Ping Tests and ARP/MAC Tables



(a) PC9 (Hub side, 192.168.2.7) pings PC6 (Switch side, 192.168.2.4): successful with ARP table showing the learned MAC address.



(a) Switch2 MAC address table after cross-LAN pings: entries learned on Fa0/1 (from hub via crossover) and Fa0/4 (directly connected host).

Figure 13: Cross-LAN connectivity verification and switch MAC address table state.

5.3.1 ARP Table of PC9 (After Cross-LAN Ping)

Internet Address	Physical Address	Type
192.168.2.4	0002.1608.0819	dynamic

Table 7: ARP table of PC9 after pinging PC6 across the hub-switch bridge.

5.3.2 MAC Address Table of Switch2 (After Cross-LAN Pings)

VLAN	MAC Address	Type	Port	Origin
1	0002.1608.0819	DYNAMIC	Fa0/1	PC6 (via switch port)
1	00e0.8f4a.4aa4	DYNAMIC	Fa0/4	PC9 (via hub crossover link)

Table 8: Switch2 MAC address table after cross-LAN communication.

Key Observation: When the hub and switch are connected via a crossover cable and all hosts are on the same subnet, communication is possible across both LAN segments at Layer 2. The switch learns the MAC addresses of hub-side hosts through the crossover link port (Fa0/4), while the hub simply broadcasts all frames received from the switch to all its connected hosts.

6 Question 5: Inter-LAN Routing with Static Routes

6.1 Problem Statement

Create a LAN (named JU-Main) with three hosts connected via a layer-2 switch (Cisco 2950 switch). Connect the switch to a router (Cisco 1841). Assign IP addresses from network 192.168.148.0/24. Create another LAN (named JU-SL) with three hosts connected via another switch and router. Assign IP addresses from network 192.168.149.0/24. Connect the two routers through WAN interfaces from network 192.168.150.0/24. Add static routes in both routers to route packets between two LANs.

6.2 Network Topology

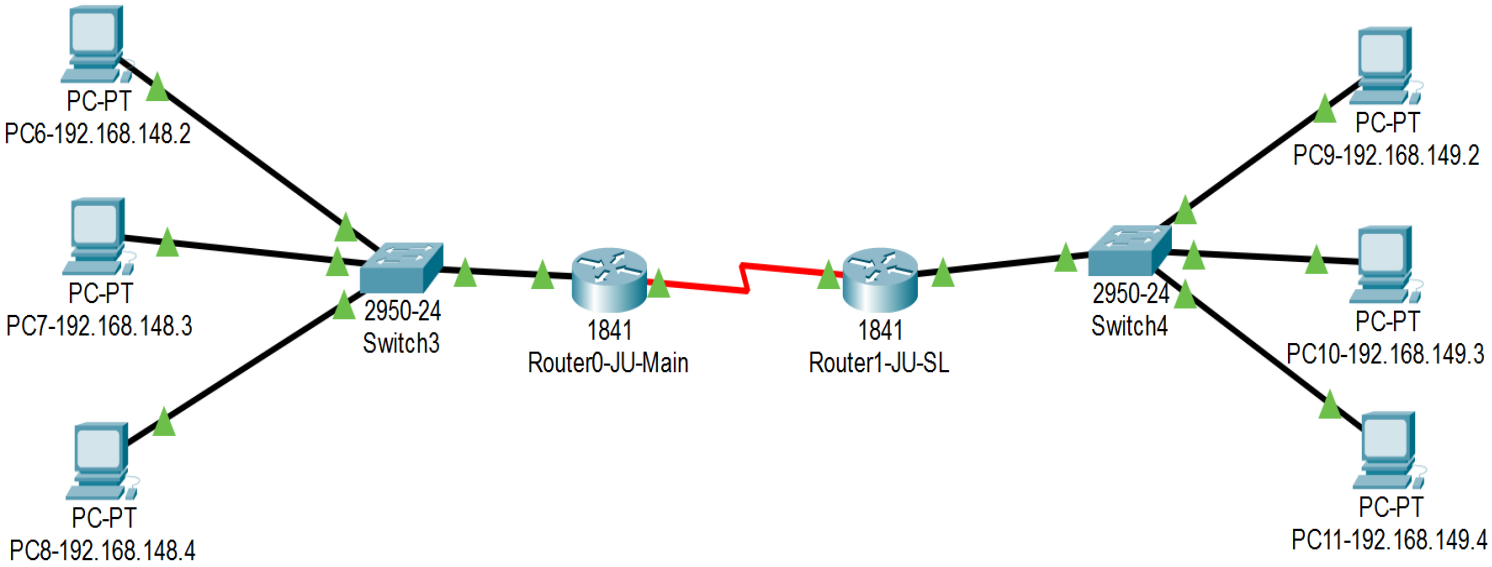


Figure 14: Complete internetwork topology: JU-Main LAN (left) and JU-SL LAN (right) interconnected via two Cisco 1841 routers (Router0-JU-Main and Router1-JU-SL) through a Serial WAN link.

6.2.1 IP Address Configuration

Network	Device	IP Address	Subnet Mask	Role
JU-Main (192.168.148.0/24)	Router0 Fa0/0	192.168.148.1	255.255.255.0	Default Gateway
	PC6	192.168.148.2	255.255.255.0	End Host
	PC7	192.168.148.3	255.255.255.0	End Host
	PC8	192.168.148.4	255.255.255.0	End Host
JU-SL (192.168.149.0/24)	Router1 Fa0/0	192.168.149.1	255.255.255.0	Default Gateway
	PC9	192.168.149.2	255.255.255.0	End Host
	PC10	192.168.149.3	255.255.255.0	End Host
	PC11	192.168.149.4	255.255.255.0	End Host
WAN Link (192.168.150.0/24)	Router0 Serial0/0/0	192.168.150.1	255.255.255.0	DCE/DTE
	Router1 Serial0/0/0	192.168.150.2	255.255.255.0	DCE/DTE

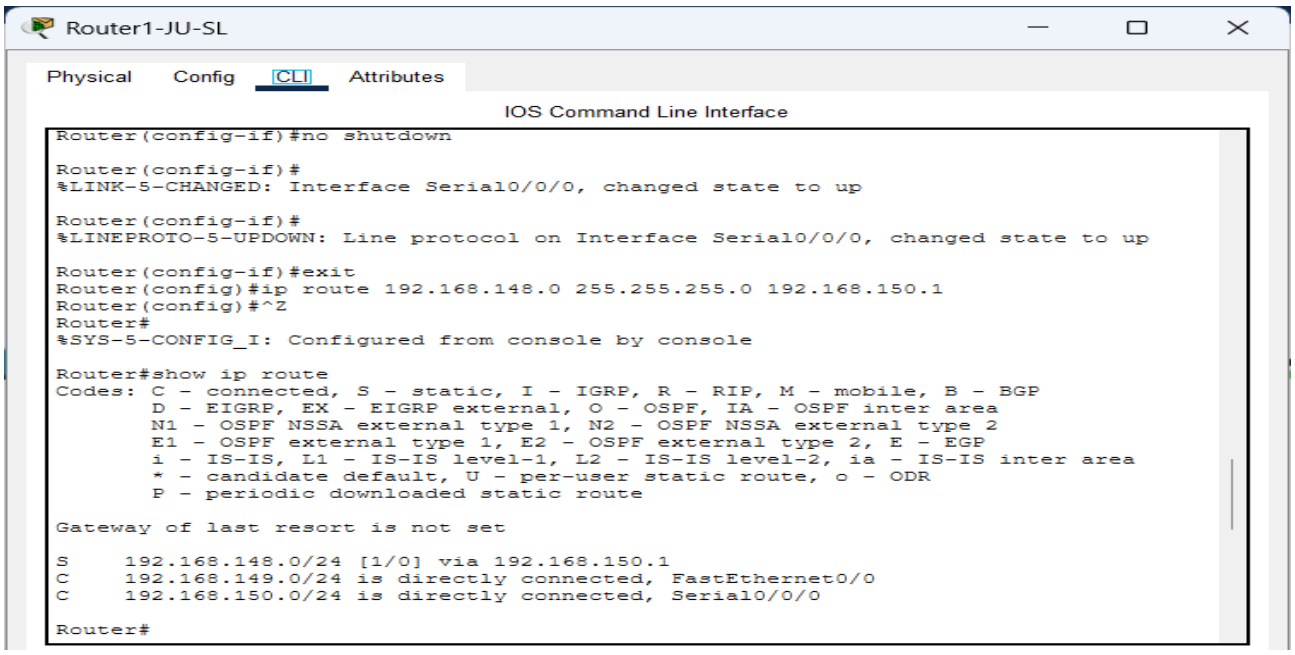
Table 9: Complete IP addressing scheme for the JU-Main and JU-SL internetwork.

6.3 Router Configuration and Static Routes

Static routes were configured on both routers to enable inter-LAN communication:



(a) Router0 (JU-Main): Routing table showing connected routes (C) for `192.168.148.0/24` and `192.168.150.0/24`, and a static route (S) to `192.168.149.0/24` via `192.168.150.2`.



(a) Router1 (JU-SL): Routing table showing connected routes (C) for `192.168.149.0/24` and `192.168.150.0/24`, and a static route (S) to `192.168.148.0/24` via `192.168.150.1`.

Figure 16: Routing tables of both routers confirming static route configuration.

6.3.1 Static Route Commands

```
! Router0-JU-Main:
Router(config)#ip route 192.168.149.0 255.255.255.0 192.168.150.2

! Router1-JU-SL:
Router(config)#ip route 192.168.148.0 255.255.255.0 192.168.150.1
```

6.4 Inter-LAN Ping Verification

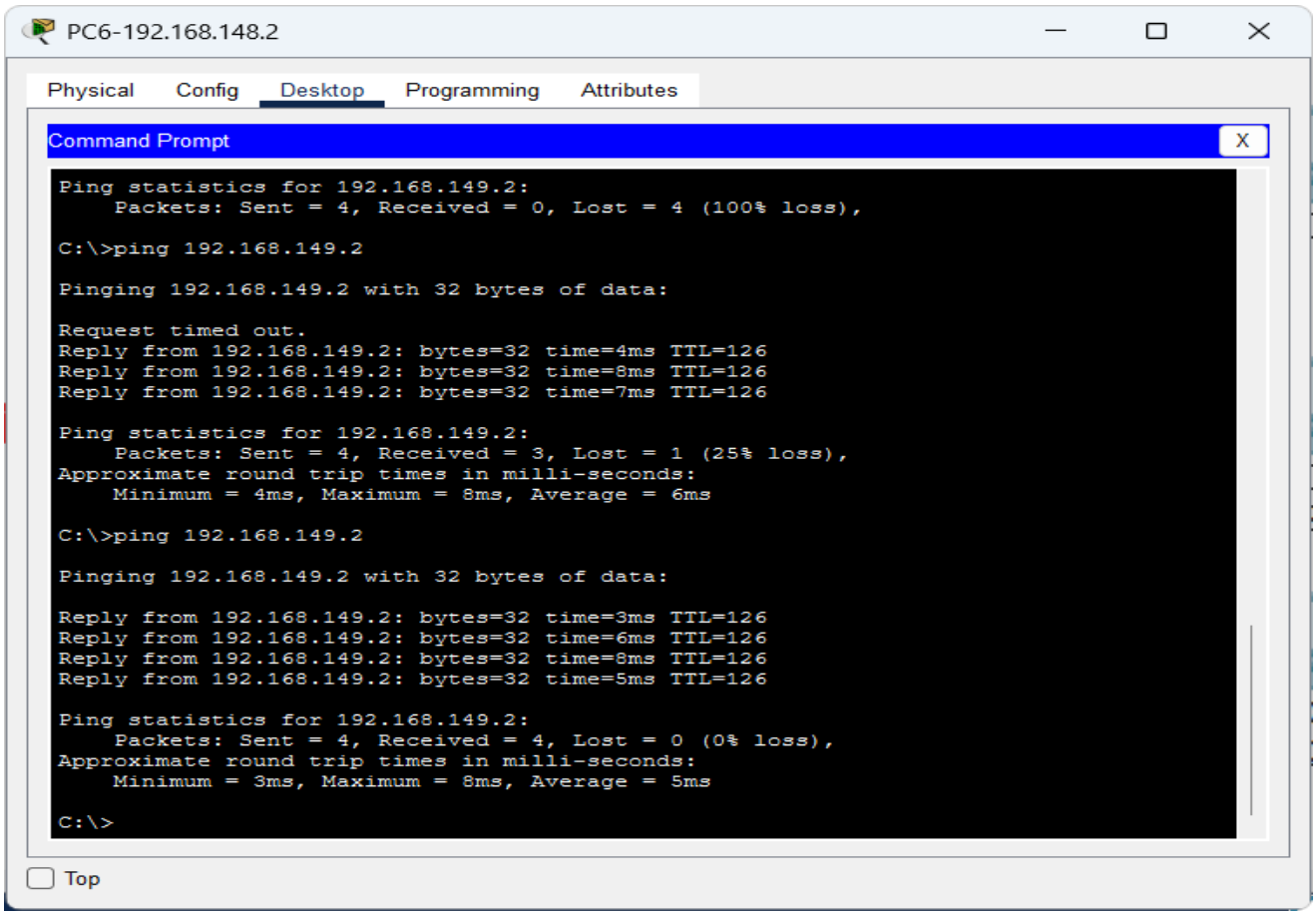


Figure 17: PC6 (`192.168.148.2`) in JU-Main pinging PC9 (`192.168.149.2`) in JU-SL: initial request times out (ARP resolution), subsequent packets successful with TTL=126 (two router hops). Second ping attempt shows 0% loss.

Key Observations:

- The first ping attempt had 25% packet loss (1 timeout) because the initial ARP resolution across routers introduces additional delay.
- The TTL value of **126** (instead of the default 128) confirms that the ICMP packet traversed **two router hops**: Router0 → WAN link → Router1.
- The second ping attempt succeeded with 0% loss, confirming that the static routes are properly configured and operational.
- Each host's default gateway must be set to the local router's interface IP address for inter-LAN routing to function correctly.

7 Question 6: DHCP Server Configuration

7.1 Problem Statement

Add servers to the individual LANs (in problem 5) and configure them as a DHCP server. Configure the hosts in the individual LAN to obtain IP addresses and address of the default gateway via this DHCP server.

7.2 Network Topology with DHCP Servers

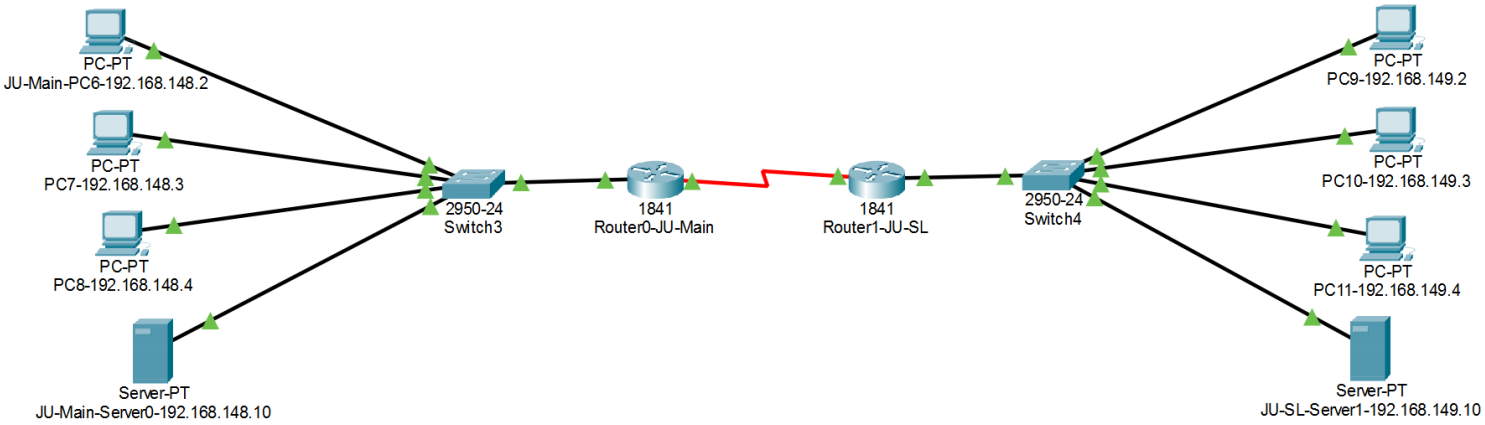


Figure 18: Extended topology from Question 5 with DHCP servers added: JU-Main-Server0 (`192.168.148.10`) on the left and JU-SL-Server1 (`192.168.149.10`) on the right.

7.3 DHCP Server Configuration

Each server was configured with the DHCP service to dynamically assign IP addresses to hosts on its respective LAN.

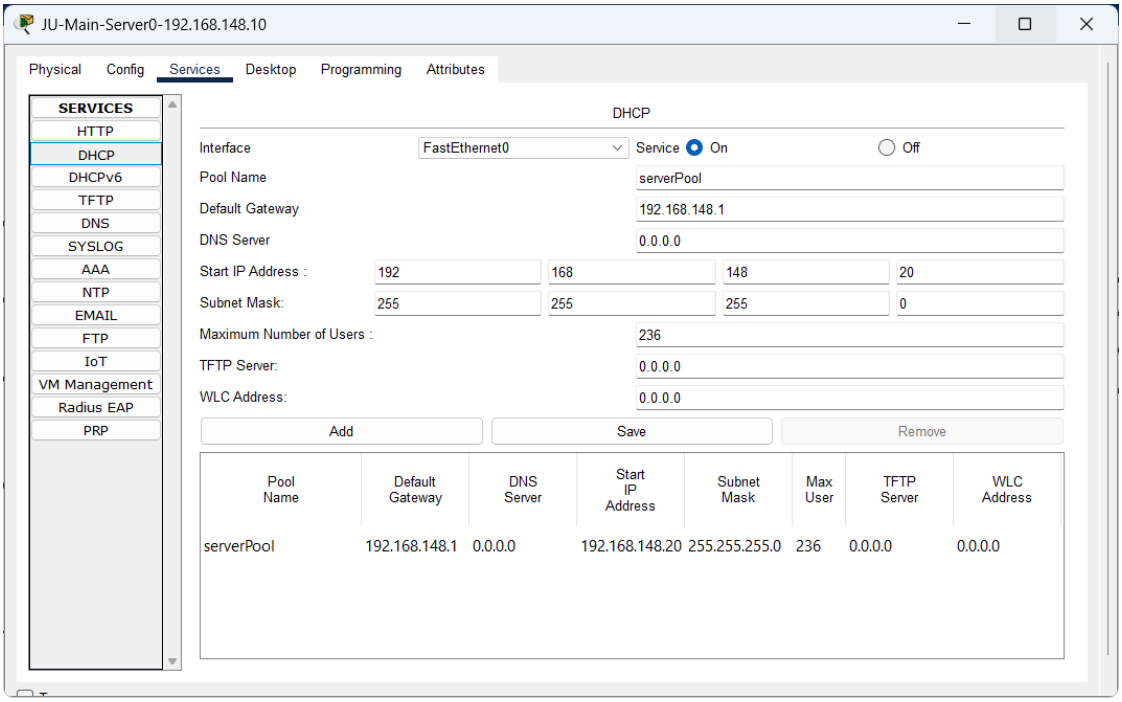


Figure 19: DHCP configuration on JU-Main-Server0: Pool name “serverPool”, Default Gateway `192.168.148.1`, Start IP Address `192.168.148.20`, Subnet Mask `255.255.255.0`, Maximum 236 users.

7.3.1 DHCP Pool Parameters

Server	Pool Name	Gateway	Start IP	Subnet Mask	Max Users
JU-Main-Server0	serverPool	192.168.148.1	192.168.148.20	255.255.255.0	236
JU-SL-Server1	serverPool	192.168.149.1	192.168.149.20	255.255.255.0	236

Table 10: DHCP pool configuration for both LAN servers.

7.4 DHCP Client Verification

Hosts were reconfigured from static to DHCP addressing mode. Upon enabling DHCP, each host broadcasts a DHCP Discover message, receives a DHCP Offer from the server, requests the offered address (DHCP Request), and receives confirmation (DHCP Acknowledge).

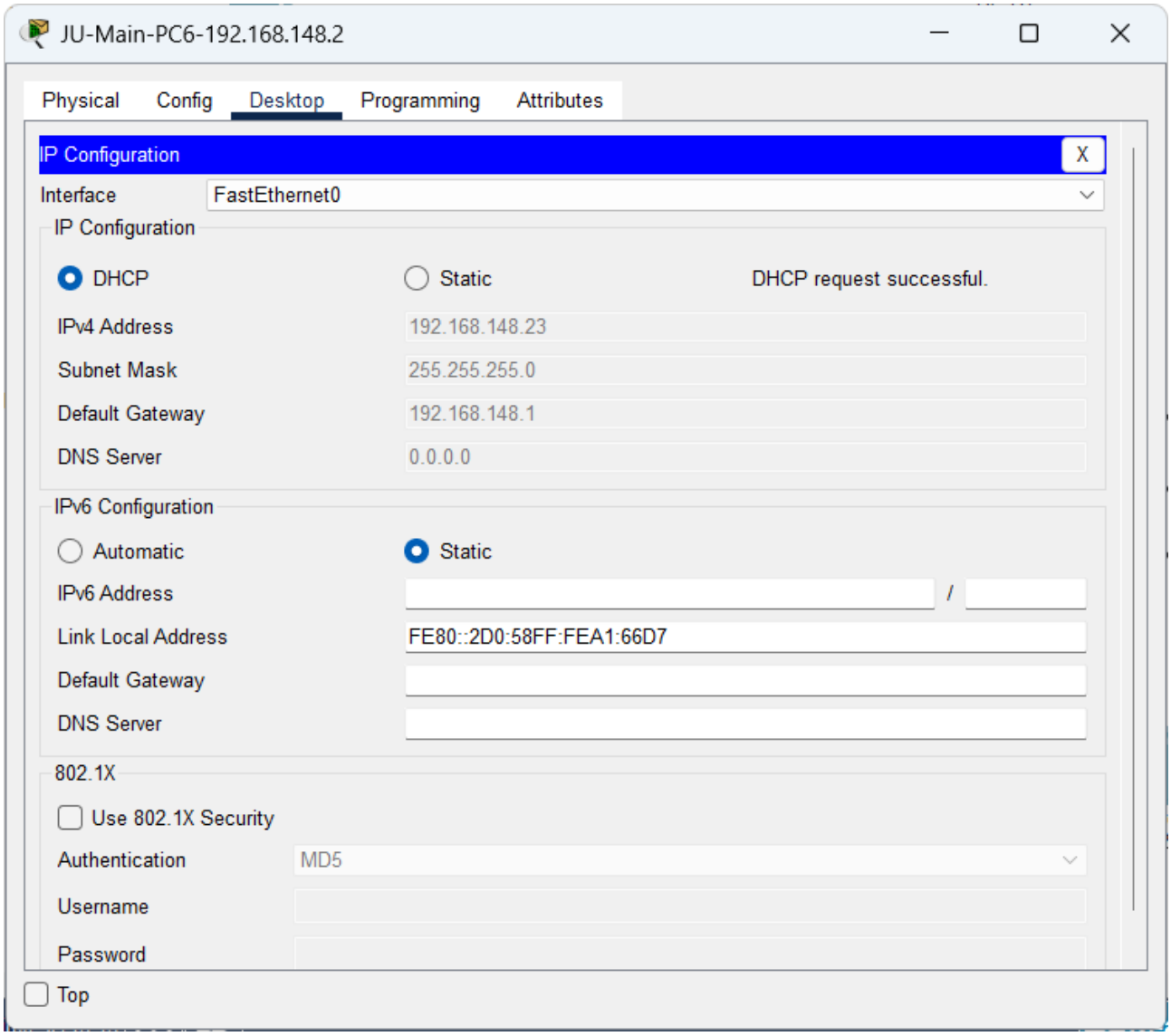


Figure 20: JU-Main-PC6 IP Configuration showing "DHCP request successful": dynamically assigned IP `192.168.148.23`, Subnet Mask `255.255.255.0`, Default Gateway `192.168.148.1`.

Key Observations:

- The DHCP server automatically assigned IP address `192.168.148.23` to PC6, starting from the configured pool start address (`192.168.148.20`).
- The default gateway (`192.168.148.1`) was automatically provided to the client, eliminating manual gateway configuration.
- The DHCP process follows the standard **DORA** sequence: **D**iscover → **O**ffer → **R**equest → **A**cknowledge.

8 Question 7: Complete LAN with DNS, Web, FTP, and DHCP Servers

8.1 Problem Statement

Create a LAN (CSE) with three hosts connected via a layer-2 switch (Cisco 2950 switch CSE-Switch). Also add a web server and an FTP server to this LAN. The hosts dynamically get their IP addresses from a local DHCP server. Servers are assigned fixed IP addresses. Add a Domain Name Server (DNS) to this LAN. Create appropriate records in the DNS server for the individual servers in the LAN. The domain name of the LAN is cse.myuniv.edu. Configure the individual hosts to use the local DNS server for name resolution.

8.2 Network Topology

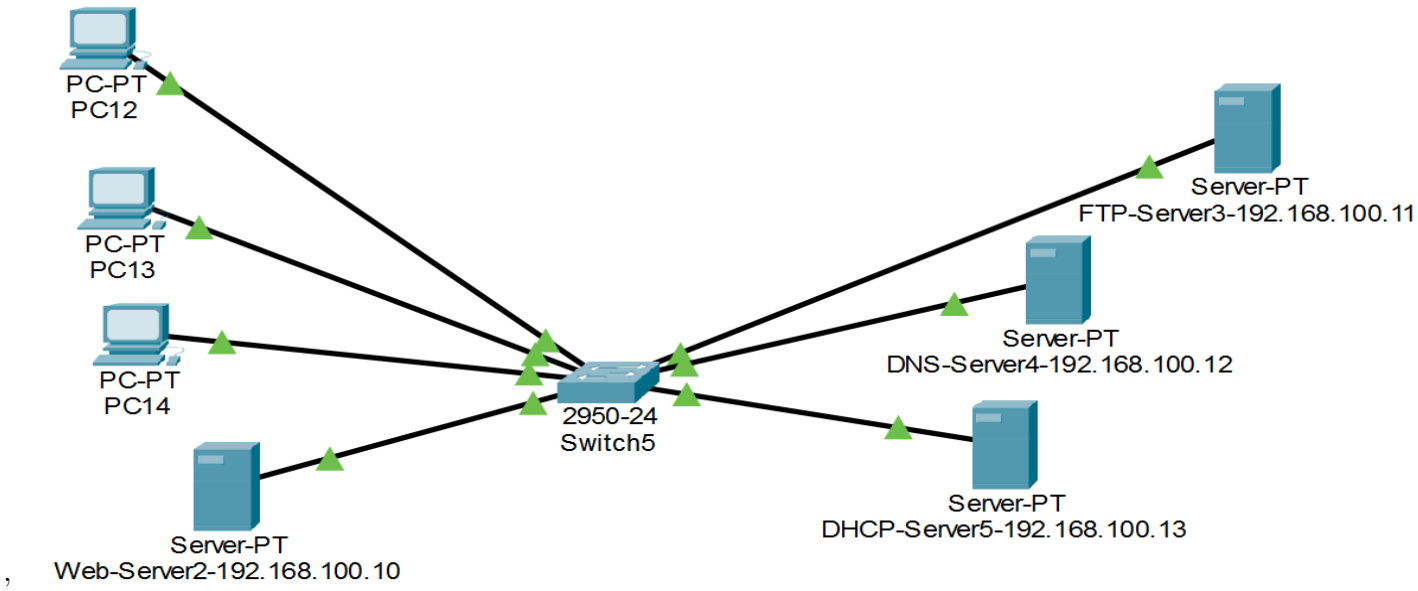


Figure 21: CSE LAN topology: Three hosts (PC12, PC13, PC14) and four servers (Web-Server2, FTP-Server3, DNS-Server4, DHCP-Server5) connected via Cisco 2950-24 Switch5.

8.2.1 Server IP Address Configuration

Server	IP Address	Subnet Mask	Service
Web-Server2	`192.168.100.10`	`255.255.255.0`	HTTP
FTP-Server3	`192.168.100.11`	`255.255.255.0`	FTP
DNS-Server4	`192.168.100.12`	`255.255.255.0`	DNS
DHCP-Server5	`192.168.100.13`	`255.255.255.0`	DHCP

Table 11: Static IP address assignment for all servers in the CSE LAN.

8.3 DNS Server Configuration

The DNS server was configured with A records mapping domain names to server IP addresses within the `cse.myuniv.edu` domain:

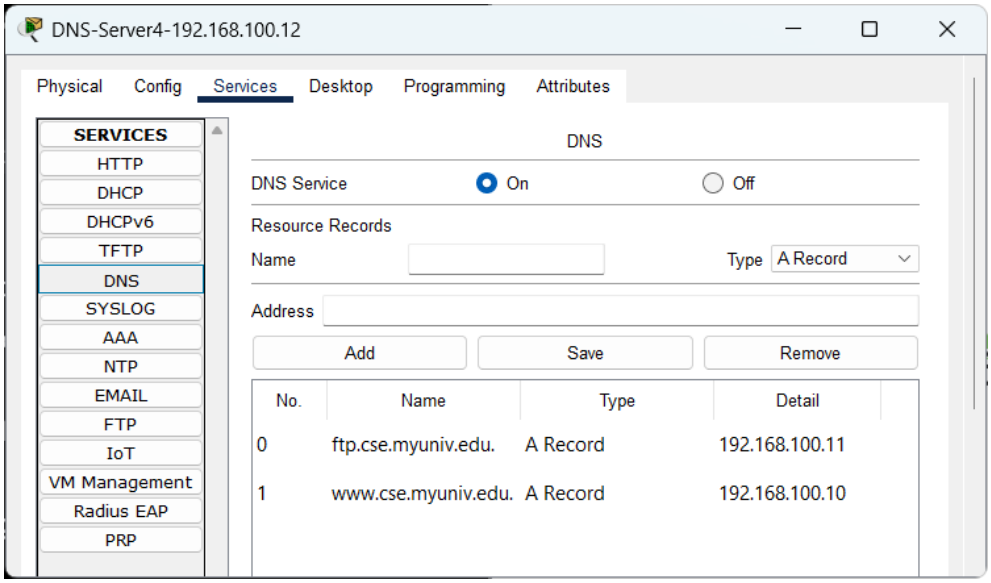


Figure 22: DNS-Server4 configuration showing two A records: `ftp.cse.myuniv.edu.` → `192.168.100.11` and `www.cse.myuniv.edu.` → `192.168.100.10`.

8.3.1 DNS Resource Records

No.	Name (FQDN)	Type	Detail (IP Address)
0	`ftp.cse.myuniv.edu.`	A Record	`192.168.100.11`
1	`www.cse.myuniv.edu.`	A Record	`192.168.100.10`

Table 12: DNS A Records configured on DNS-Server4 for the `cse.myuniv.edu` domain.

8.4 Web Server Access via DNS

With DNS properly configured, hosts can access the web server using its domain name rather than its IP address.

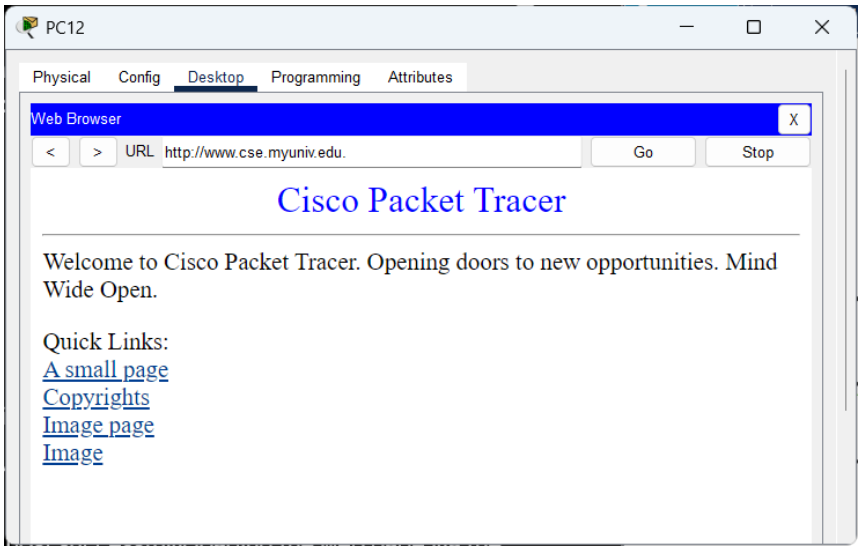


Figure 23: PC12 accessing the web server via URL `http://www.cse.myuniv.edu.`: the DNS server resolves the domain name to `192.168.100.10`, and the Cisco Packet Tracer default web page is displayed.

8.5 FTP Server Access via DNS

The FTP server was also accessed using its DNS-resolved domain name:

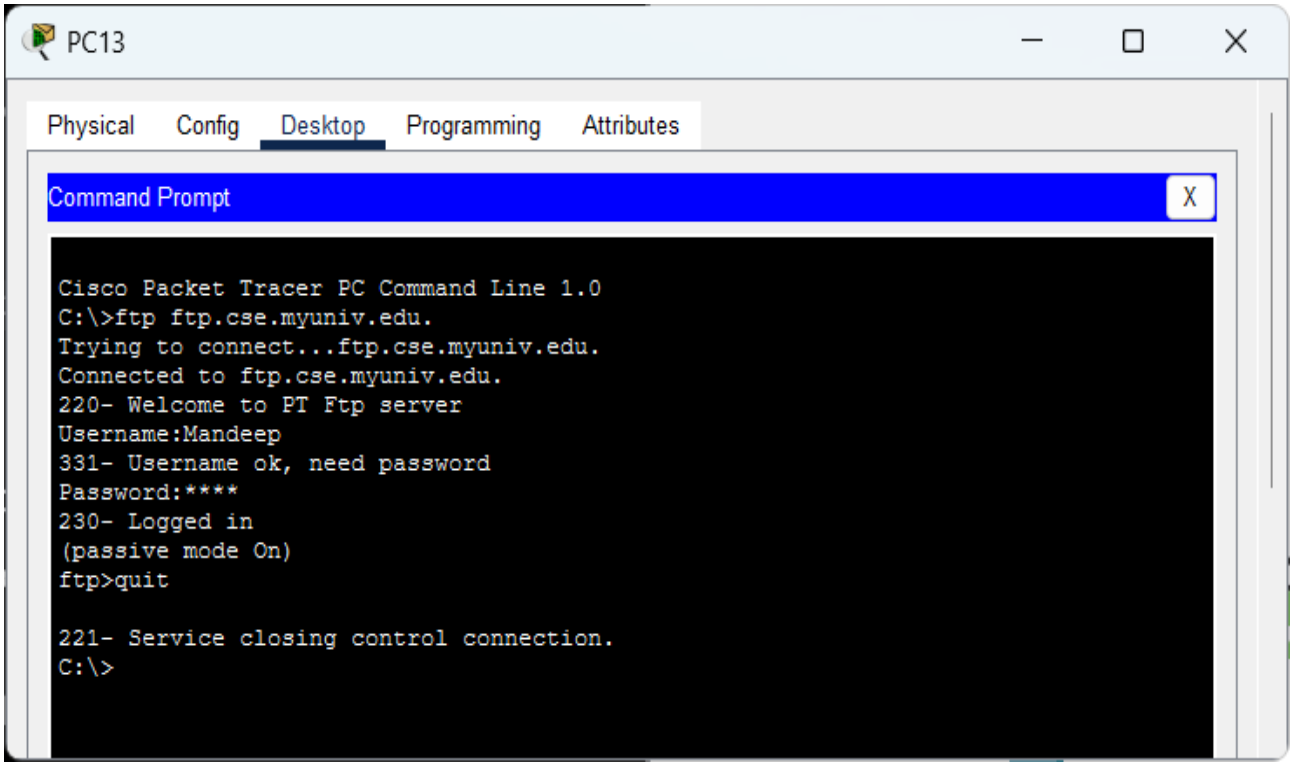


Figure 24: PC13 connecting to the FTP server via `ftp ftp.cse.myuniv.edu.`: the DNS server resolves the domain name, user “Mandeep” authenticates successfully, and the session is established.

As shown in Figure 24, the FTP session proceeds through the following steps:

1. DNS resolution: `ftp.cse.myuniv.edu.` → `192.168.100.11`
2. TCP connection established to port 21
3. Server banner: `220- Welcome to PT FTP server`
4. User authentication: Username "Mandeep" with password → `230- Logged in`
5. Passive mode enabled, session operational
6. Clean disconnection via `quit` → `221- Service closing control connection`

Key Observations:

- The DHCP server dynamically assigns IP addresses and DNS server addresses to the three hosts.
- The DNS server provides centralized name resolution, allowing hosts to access services by memorable domain names instead of IP addresses.
- Both HTTP and FTP services are fully operational and accessible through DNS-resolved FQDNs.
- The domain structure follows the convention: `service.cse.myuniv.edu`, where `service` is either `www` or `ftp`.

9 Conclusion and Comments

9.1 Summary of Results

This laboratory assignment successfully demonstrated progressively complex network configurations using Cisco Packet Tracer. Key findings across all seven experiments include:

1. **Direct Connectivity:** Two PCs connected via a crossover cable can communicate directly without any intermediate networking device, provided they are on the same IP subnet.
2. **Hub-based LAN:** A hub creates a single broadcast and collision domain. All connected hosts can communicate, but all traffic is visible to all hosts (no frame filtering).
3. **Switch-based LAN:** A switch dynamically learns MAC addresses and builds a forwarding table, enabling intelligent unicast forwarding. Before any traffic, both ARP and MAC tables are empty; after pinging, they are fully populated.
4. **Hub-Switch Interconnection:** Hosts on a hub and switch can communicate when connected via a crossover cable on the same subnet. The switch learns hub-side MAC addresses through the interconnection port.
5. **Inter-LAN Routing:** Static routes enable communication between hosts on different subnets across router boundaries. The TTL decrement (128 → 126) confirms two-hop routing.
6. **DHCP Automation:** DHCP servers eliminate manual IP configuration by dynamically assigning addresses, subnet masks, and default gateways to clients via the DORA process.
7. **DNS-based Service Access:** DNS servers provide domain-to-IP resolution, enabling user-friendly access to web and FTP services using fully qualified domain names.

9.2 What I Learned

- Understood the operational differences between Layer-1 hubs (broadcast) and Layer-2 switches (intelligent forwarding).
- Gained practical experience in configuring IP addressing, static routing, DHCP, and DNS services.
- Observed the relationship between ARP tables (IP-to-MAC mapping) and switch MAC address tables (MAC-to-port mapping).
- Learned to configure WAN serial links between routers and establish inter-LAN reachability through static routes.

9.3 References Used

- Kurose, J. F., and Ross, K. W., *Computer Networking: A Top-Down Approach*, 8th ed., Pearson, 2021.
- Cisco Systems, *Cisco Packet Tracer Documentation*, https://www.netacad.com/courses/packet-tracer, 2026.
- IEEE Standards Association, *IEEE Standard for Ethernet*, IEEE Std 802.3-2022.